

Department of Mechanical Engineering, NIT Kurukshetra

B. Tech. 3rd Sem, Production and Industrial Engg. (PIE) - (Date: 25-09-2025)

Facilities Design (PIPC-203)

Mid-Term Exam-1

Max. Marks:15

Time: -- 50 Min

Q1: Select the facility based on *composite measure method* for the data received from the location analyst. The three potential locations with five relevant factors like transportation cost per week, labour costs per week, finishing material supply, maintenance facilities and community attitude. The costs are in rupees whereas for the last three factors, points are assigned on 0-100 scale. The payoff matrix is given in following table.

Factors		Potential location sites		
		L1	L2	L3
Transportation cost / week (Rs.)	F1	930	860	750
Labour cost / week (Rs.)	F2	1340	1160	1250
Finishing Materials Supply	F3	36	76	78
Maintenance Facility	F4	69	35	31
Community Attitude	F5	59	89	79

Company management has pre-established weights in various factors and they include a standard of 1.75 for each Rs. 20 per week of economic advantage. Other weight applications are: Finishing materials supply: 1.35,

Maintenance Facility : 0.8, Community Attitude : 2.6,

Maximum acceptable score for maintenance facility is 36 and minimum acceptable Materials Supply score in 100. Suggest the suitable location. (6 marks)

Q2: Calculate the machine requirement in product layout for the new facility in given below context. Product 'X' has been sale forecasted for 46,350 units/yr and the one shift for 332 days is planned for production. The operations 'a', 'b', and 'c' are required on machine 'A', 'B' and 'C' with standard time of operation of 9 min/unit, 6 min/unit and 11 min per unit respectively. In optimistic way, the production recovery is expected to be 87%. (5 marks)

Q3: Elaborate in brief the qualitative methods of factor rating giving relevant examples. (4 marks)